

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

5th Semester of B. Tech. Examination (EE)

Nov-Dec 2012

EE 302 ELCTRICAL POWER SYSTEM-II

Date: 29.11.2012, Thursday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Explain the Choice and Spacing of conductor for High voltage transmission line. [07]
- Q - 2 (a) Write Short note on Radio Interference. [07]
- (b) Explain the Theory of Corona formation. [05]
- (c) Explain the following term: [02]
- (1) Visual Critical Voltage (2) Critical disruptive Voltage

OR

- Q - 2 (a) Write Short note on Insulation Coordination. [07]
- (b) Write technical note on Surge Impedance Loading of transmission line. [07]
- Q - 3 (a) Explain the Selection of size and Location of generating stations. [06]
- (b) Write short note on Lamp Flicker. [06]
- (c) Give the advantages of Interconnection. [02]

OR

- Q - 3 (a) List out main equipments of substation and explain the functions of each [06]
- (b) What is Interconnector ? Explain the Interconnected system with neat sketch. [04]
- (c) A 2-Wire d.c. distributor AB 500 metres long is fed from both ends and is loaded uniformly at the rate of 1.0 A/ metre. At feeding point A, the voltage is maintained at 255 V and at B 250 V. If the resistance of each conductor is 0.1Ω per kilometer, determine: [04]

- (i) The minimum voltage and the point where it occurs
- (ii) The currents supplied from feeding points A and B.

SECTION - II

- Q - 4 (a) Explain the Kelvin's Law with its Limitation. [07]
- Q - 5 (a) Explain distribution system planning model with its block diagram. [07]
- (b) A d.c ring main ABCDA is fed from point A from a 250 V supply and the resistance (including both lead and return) of various sections are as follows: $AB=0.02 \Omega$; $BC=0.018 \Omega$; $CD=0.025 \Omega$ and $DA=0.02 \Omega$. The main supplies loads of 150 A at B; 300 A at C and 250 A at D. determine the voltage at each load point. If the points A and C are linked through an interconnector of resistance 0.02Ω , determine the new voltage at each load point. [07]

OR

- Q - 5 (a) Explain the Main step for Power system planning with diagram. [06]
- (b) A single phase distributor one km long has resistance and reactance per conductor of 0.1Ω and 0.15Ω respectively. At the far end, the voltage $V_B=200$ V and the current is 100 A at a p.f. of 0.8 lagging. At the mid-point M of the distributor, a current of 100 A is tapped at a p.f. of 0.6 lagging with reference to the voltage V_M at the midpoint. Calculate : [08]
- (i) Voltage at midpoint
 - (ii) Sending end voltage V_A
 - (iii) Phase angle between V_A and V_B
- Q - 6 (a) Write short note on Single Wire Earth Return system with its advantages and limitations. [07]
- (b) The cost per km for each of the copper conductor of a section $a \text{ cm}^2$ for a transmission line is Rs $(2,800 a + 1300)$. The load factor of the load current is 80% and the load factor of the losses is 65%. The rate of interest and depreciation is 10% and the cost of energy is 50 paise per kWh. Find the most economical current density for the transmission line by the use of Kelvin's law. Given $\rho = 1.78 \times 10^{-8} \Omega\text{-m}$. [07]

OR

- Q - 6 (a) Write short note on Auto-Recloser. [07]
- (b) List out the common causes of unsatisfactory performance of rural distribution system and also suggest the remedial measures for it. [07]